Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: B.Tech Semester: III

Name of the Paper: Fluid and Particle Mechanics Paper Code: **TPE 303**

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20marks)	**CO1**
(a)	What are dimensions? Why is Renolds number dimensionless?	
(b)	Derive steady continuity equation in cartesian coordinates.	
(c)	Explain the rise of pressure with depth in a fluid.	
Q2	(20 marks)	**CO1,CO2**
(a)	How can the Bernoulli equation be modified to account for losses in a flow? What are major losses?	
(b)	How do you select variable for pi-terms in Buckingham Pi theorem?	
(c)	How many repeating variables and pi-terms should be there in Buckingham Pi theorem?	
Q3	(20 marks)	**CO2,CO3**
(a)	With diagram explain the difference between a single acting and double acting reciprocating pump.	
(b)	Explain the Reynolds number and Mach number.	
(c)	What are the different parts of centrifugal pump? On what principle does it work?	
Q4	(20 marks)	**CO3,CO4**
(a)	What is need for priming in a centrifugal pump?	
(b)	Derive the shear stress distribution for a circular pipe in a laminar flow	
(c)	What are the main components of reciprocating pump?	
Q5	(20 marks)	**CO5,CO6**
(a)	Explain Stoke's law.	
(b)	What is a notch? Explain with the help of diagram. Define the coefficient of discharge.	
(c)	What do you understand by fluidization? What are its applications?	

Note for the question paper setters:

- **Question paper should cover all the COs of the course.**